COMPUTER ENGINEERING

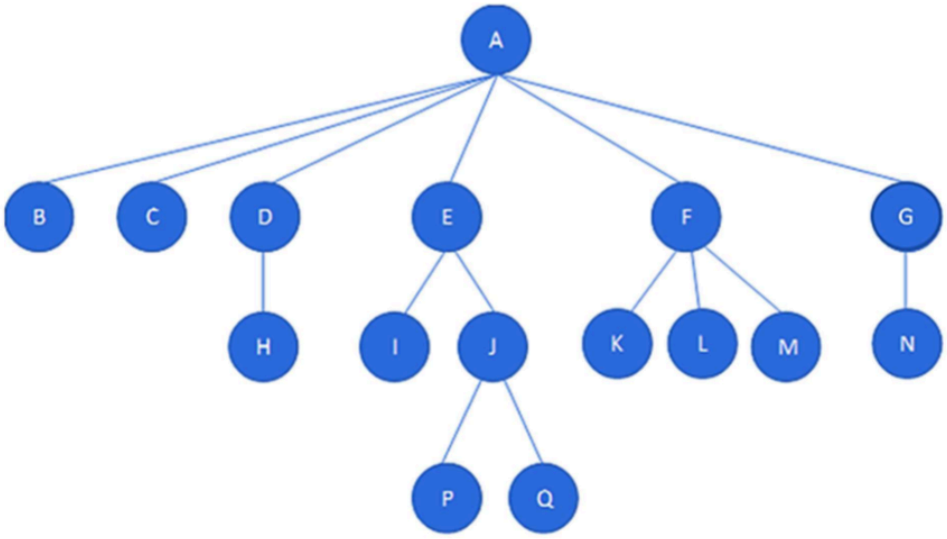
QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structures and Algorithms	**SECTION:**	CPE21S1
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ACTIVITY NO. 9.1 **Tree ADT**

1. PROCEDURE OUTPUT

ILO A: Create C++ code to implement a general tree, binary tree and binary search tree.



Tree.cpp

```
#include <iostream>
#include "Tree.h"

int main() {
    //NODE
    GeneralTree<char> A('A');

    GeneralTree<char>* B = A.addChild('B');
```

```
    GeneralTree<char>* C = A.addChild('C');
    GeneralTree<char>* D = A.addChild('D');
    GeneralTree<char>* E = A.addChild('E');
    GeneralTree<char>* F = A.addChild('F');
    GeneralTree<char>* G = A.addChild('G');

    GeneralTree<char>* H = D->addChild('H');
    GeneralTree<char>* I = E->addChild('I');
    GeneralTree<char>* J = E->addChild('J');
    GeneralTree<char>* P = J->addChild('P');
    GeneralTree<char>* Q = J->addChild('Q');
    GeneralTree<char>* K = F->addChild('K');
    GeneralTree<char>* L = F->addChild('L');
    GeneralTree<char>* M = F->addChild('M');
    GeneralTree<char>* N = G->addChild('N');


    std::cout << "\n===== Task 3.2: Traversal Output =====\n";
    std::cout << "Pre-order:  "; A.preOrder();  std::cout << "\n";
    std::cout << "Post-order: "; A.postOrder(); std::cout << "\n";
    std::cout << "In-order:   "; A.inOrder();   std::cout << "\n";

    std::cout << "\n===== Task 3.3: findData Demo =====\n";
    bool found1 = false;
    std::cout << "findData(1, 'J') -> ";
    A.findData(1, 'J', found1);
    if (!found1) std::cout << "(not found)\n";

    bool found2 = false;
    std::cout << "findData(2, 'Z') -> ";
    A.findData(2, 'Z', found2);
    if (!found2) std::cout << "(no output, key not in tree)\n";

    std::cout << "\n===== Task 3.4: Add leaf 'O' as a child of G
=====\n";
    G->addChild('O');
```

```
    std::cout << "Pre-order after insert:  "; A.preOrder();
std::cout << "\n";

    bool found3 = false;
    std::cout << "findData(1, 'O') -> ";
    A.findData(1, 'O', found3);
    if (!found3) std::cout << "(not found)\n";

    return 0;
}
```

Tree.h

```
#ifndef TREE_H
#define TREE_H

#include <iostream>

template <typename T>
class GeneralTree {
private:
    T key;
    GeneralTree<T>* firstChild;
    GeneralTree<T>* nextSibling;

public:
    GeneralTree(T val) : key(val), firstChild(nullptr),
nextSibling(nullptr) {}

    ~GeneralTree() {
        delete firstChild;
        delete nextSibling;
    }

    T getKey() {
```

```
        return key;
    }

    //Creates a new node
    GeneralTree<T>* addChild(T value) {
        GeneralTree<T>* child = new GeneralTree<T>(value);

        if (firstChild == nullptr) {
            firstChild = child;
        } else {
            GeneralTree<T>* sibling = firstChild;
            while (sibling->nextSibling != nullptr) {
                sibling = sibling->nextSibling;
            }
            sibling->nextSibling = child;
        }

        return child;
    }




    void preOrder() {
        std::cout << key << " ";

        for (GeneralTree<T>* c = firstChild; c != nullptr; c =
c->nextSibling) {
            c->preOrder();
        }
    }

    void postOrder() {
        for (GeneralTree<T>* c = firstChild; c != nullptr; c =
c->nextSibling) {
            c->postOrder();
```

```
        }

        std::cout << key << " ";
    }


    void inOrder() {
        if (firstChild != nullptr) {
            firstChild->inOrder();
        }

        std::cout << key << " ";

        if (firstChild != nullptr) {
            for (GeneralTree<T>* c = firstChild->nextSibling; c != nullptr; c = c->nextSibling) {
                c->inOrder();
            }
        }
    }

    //Locates the node holding `value` anywhere in this subtree.
    GeneralTree<T>* find(T value) {
        if (key == value) {
            return this;
        }

        for (GeneralTree<T>* c = firstChild; c != nullptr; c = c->nextSibling) {
            GeneralTree<T>* result = c->find(value);
            if (result != nullptr) {
                return result;
            }
        }

        return nullptr;
```

```
    }

    //Traverses using the chosen order
    void findData(int choice, T target, bool& foundFlag) {
        if (foundFlag) {
            return;
        }

        if (choice == 1) {
            if (key == target) {
                std::cout << target << " was found!" << std::endl;
                foundFlag = true;
            }
            for (GeneralTree<T>* c = firstChild; c != nullptr &&
!foundFlag; c = c->nextSibling) {
                c->findData(choice, target, foundFlag);
            }
        } else if (choice == 2) {
            for (GeneralTree<T>* c = firstChild; c != nullptr &&
!foundFlag; c = c->nextSibling) {
                c->findData(choice, target, foundFlag);
            }
            if (!foundFlag && key == target) {
                std::cout << target << " was found!" << std::endl;
                foundFlag = true;
            }
        } else if (choice == 3) {
            if (firstChild != nullptr) {
                firstChild->findData(choice, target, foundFlag);
            }
            if (!foundFlag && key == target) {
                std::cout << target << " was found!" << std::endl;
                foundFlag = true;
            }
            if (firstChild != nullptr) {
                for (GeneralTree<T>* c = firstChild->nextSibling; c
```

```
!= nullptr && !foundFlag; c = c->nextSibling) {
                    c->findData(choice, target, foundFlag);
                }
            }
        }
    }
};

#endif
```

OUTPUT

```
===== Task 3.2: Traversal Output =====
Pre-order:  A B C D H E I J P Q F K L M G N
Post-order: B C H D I P Q J E K L M F N G A
In-order:   B A C H D I E P J Q K F L M N G

===== Task 3.3: findData Demo =====
findData(1, 'J') -> J was found!
findData(2, 'Z') -> (no output, key not in tree)

===== Task 3.4: Add leaf 'O' as a child of G =====
Pre-order after insert:  A B C D H E I J P Q F K L M G N O
findData(1, 'O') -> O was found!
```

Node	Height	Depth
A	3	0
B	0	1
C	0	1
D	1	1
E	2	1
F	1	1

G	1	1
H	0	2
I	0	2
J	1	2
K	0	2
L	0	2
M	0	2
N	0	2
P	0	3
Q	0	3

Table 9.2

ILO B: Create C++ code for implementation of tree traversal methods such as pre-order, in-order and post- order traversal

Task 3: After implementing the code for above, answer the following:

3.1 Given the tree diagram, find the result of the pre-order, post-order and in-order traversal strategies by hand. Include this as table 9-3 in section 6.

Traversal	**Result**
Pre-order	A, B, C, D, H, E, I, J, P, Q, F, K, L, M, G, N
Post-order	B, C, H, D, I, P, Q, J, E, K, L, M, F, N, G, A
In-order	B, A, C, H, D, I, E, P, J, Q, K, F, L, M, N, G

Table 9.3

3.2 Create a function for pre-order, post-order and in-order traversal. Make sure that each function displays an output into the console. Once you have the output, create and fill table 9-4 in section 6 so that it contains the screenshot of the function, screenshot of the output and your observations. Your observations consist of a comparison between output in #1 and the output of your functions in #2.

```
45      void preOrder() {
46          std::cout << key << " ";
47
48          for (GeneralTree<T>* c = firstChild; c != nullptr; c = c->nextSibling) {
49              c->preOrder();
50          }
51      }
52
53      void postOrder() {
54          for (GeneralTree<T>* c = firstChild; c != nullptr; c = c->nextSibling) {
55              c->postOrder();
56          }
57
58          std::cout << key << " ";
59      }
60
61
62      void inOrder() {
63          if (firstChild != nullptr) {
64              firstChild->inOrder();
65          }
66
67          std::cout << key << " ";
68
69          if (firstChild != nullptr) {
70              for (GeneralTree<T>* c = firstChild->nextSibling; c != nullptr; c = c->nextSibling) {
71                  c->inOrder();
72              }
73          }
74      }
```

Table 9.4

3.3. Create a function called findData that will take the following parameter: “CHOICE, KEY”. Choice will determine which traversal method will be used to traverse the tree to find the second parameter KEY. If the KEY is found, the value will be displayed such that: “{KEY} was found!”. Otherwise, no output. Include your output as table 9-5 in section 6.

```
    //Locates the node holding `value` anywhere in this subtree.
    GeneralTree<T>* find(T value) {
        if (key == value) {
            return this;
        }

        for (GeneralTree<T>* c = firstChild; c != nullptr; c = c->nextSibling) {
            GeneralTree<T>* result = c->find(value);
            if (result != nullptr) {
                return result;
            }
        }

        return nullptr;
    }

    //Traverses using the chosen order
    void findData(int choice, T target, bool& foundFlag) {
        if (foundFlag) {
            return;
        }

        if (choice == 1) {
            if (key == target) {
                std::cout << target << " was found!" << std::endl;
                foundFlag = true;
            }
            for (GeneralTree<T>* c = firstChild; c != nullptr && !foundFlag; c = c->nextSibling) {
                c->findData(choice, target, foundFlag);
            }
        } else if (choice == 2) {
            for (GeneralTree<T>* c = firstChild; c != nullptr && !foundFlag; c = c->nextSibling) {
                c->findData(choice, target, foundFlag);
            }
            if (!foundFlag && key == target) {
                std::cout << target << " was found!" << std::endl;
                foundFlag = true;
            }
        } else if (choice == 3) {
            if (firstChild != nullptr) {
                firstChild->findData(choice, target, foundFlag);
            }
            if (!foundFlag && key == target) {
                std::cout << target << " was found!" << std::endl;
                foundFlag = true;
            }
            if (firstChild != nullptr) {
                for (GeneralTree<T>* c = firstChild->nextSibling; c != nullptr && !foundFlag; c = c->nextSibli
                    c->findData(choice, target, foundFlag);
                }
            }
        }
    }
};

#endif
```

Table 9.5

3.4. Add a new leaf node called which will contain the data value of “O”. Make this node the child of nodeG. Perform the findData function you created in #3, what can you observe about the output? Include your output and answers as table 9-6 in section 6

```
G->addChild('O');
std::cout << "Pre-order after insert:  "; A.preOrder(); std::cout << "\n";
```

Observation: the traversal correctly picks up the newly inserted node without any other code changes — because pre/post/in-order all walk firstChild → nextSibling pointers generically, adding a node anywhere in the tree is automatically included in every traversal the next time it runs. Its position in the output (right after G) reflects exactly where it was attached.

Table 9.6

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve given problems using the tree data structure’s implementation in C++

BST.cpp

```
#include <iostream>
#include "BST.h"

int main() {
    // NODE
    BinarySearchTree<int> tree(2);

    tree.insert(3);
    tree.insert(9);
    tree.insert(18);
    tree.insert(0);
    tree.insert(1);
    tree.insert(4);
    tree.insert(5);

    std::cout << "Pre-order:  "; tree.preOrder();  std::cout << "\n";
    std::cout << "In-order:   "; tree.inOrder();   std::cout << "\n";
```

```
    std::cout << "Post-order: "; tree.postOrder(); std::cout << "\n";

    return 0;
}
```

BST.h

```
#ifndef BST_H
#define BST_H

#include <iostream>

template <typename T>
class BinarySearchTree {
private:
    T key;
    BinarySearchTree<T>* left;
    BinarySearchTree<T>* right;

public:
    BinarySearchTree(T val) : key(val), left(nullptr), right(nullptr)
{}

    ~BinarySearchTree() {
        delete left;
        delete right;
    }

    void insert(T value) {
        if (value < key) {
            if (left == nullptr) {
                left = new BinarySearchTree<T>(value);
            } else {
                left->insert(value);
            }
        } else if (value > key) {
            if (right == nullptr) {
```

```
                right = new BinarySearchTree<T>(value);
            } else {
                right->insert(value);
            }
        }
    }

    void preOrder() {
        std::cout << key << " ";

        if (left != nullptr) {
            left->preOrder();
        }

        if (right != nullptr) {
            right->preOrder();
        }
    }

    void inOrder() {
        if (left != nullptr) {
            left->inOrder();
        }

        std::cout << key << " ";

        if (right != nullptr) {
            right->inOrder();
        }
    }

    void postOrder() {
        if (left != nullptr) {
            left->postOrder();
        }

        if (right != nullptr) {
```

```
            right->postOrder();
        }

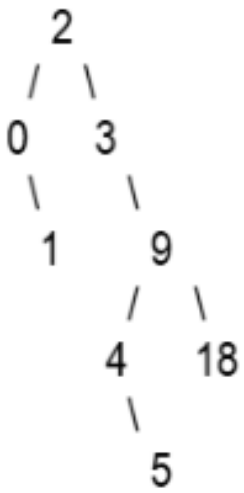
        std::cout << key << " ";
    }
};

#endif
```

OUTPUT

```
E21S1\HoA9p1\" ; if ($?) { g++ BST.cpp -o B
Pre-order:  2 0 1 3 9 4 5 18
In-order:   0 1 2 3 4 5 9 18
Post-order: 1 0 5 4 18 9 3 2
PS C:\Users\Benj Federic De Leon\Documents\
```

Structure



Traversal	Visit order
In-order	0 → 1 → 2 → 3 → 4 → 5 → 9 → 18
Pre-order	2 → 0 → 1 → 3 → 9 → 4 → 5 → 18

Post-order	1 → 0 → 5 → 4 → 18 → 9 → 3 → 2

Pre-order: 2 0 1 3 9 4 5 18
In-order: 0 1 2 3 4 5 9 18
Post-order: 1 0 5 4 18 9 3 2

Because this is a proper Binary Search Tree (BST), an in-order traversal always lists the numbers in sorted, ascending order (0, 1, 2, 3, 4, 5, 9, 18), which is the defining rule of a BST. In contrast, pre-order and post-order traversals do not sort the numbers, but instead show how the tree is shaped. Specifically, pre-order visits parents before children to help copy or save the tree, while post-order visits children before parents to help safely delete the tree from the bottom up.

Code analysis

The program basically creates a Binary Search Tree with the 2 as the root node and inserts the values then smaller values will then be placed on the left and the larger values on the right. First is the insert() functions it basically finds the correct position of each value. After that it uses preOrder(), inOrder and postOrder to display the tree in different orders after that the destructor deletes the left and right nodes.

Conclusion

This activity helped me understand Binary Search Trees (BST), a non-linear data structure that relies on linked lists but does not follow a single straight path like stacks or queues. I learned essential tree components like roots, keys, and children, and utilize three traversal methods: In-order, Pre-order, and Post-order. While I successfully grasped how BST concepts and methods work, I need to practice hard-coding more to reduce my reliance on references and examples in the future.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: Gemini, Grammarly

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: *Code assistance, explanation of concepts, and summarization, grammar checking*

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: Gemini was used to find/fix some bugs and guide us through the supplementary code, but we reviewed and understood it.

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

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